

Joseph Zaki

October 4, 2026

COM SCI 910.3

Assignment 1 Reflection

In my first implementation of the agents and tasks, I began by giving clear, concise instructions in the task descriptions, roles, and backstories. I initially under-utilized the “expected_output” field in the task definitions. I found that without specific file types and structures defined for the output the agents produced unreliable results. For instance, the search task would sometimes return sources without the relevant passages or passages without the sources. At times the result would be plain text, other times it would be markdown. Early on I specified that this task should produce a JSON file with a key for each sub-question created by the “expand_task” agent. This led to the most impactful results throughout my process of building the four-agent system. Having a consistent format for the sources and citations then allowed me to better shape the prompts for the synthesizer and report writer.

When reviewing the initial results of the synthesizer and report writer, I noticed they focused too much on the sub-questions created by the query expander agent and “forgot” to address the research question clearly. I was able to address this by directly instructing them in the task descriptions to address the research question. A recurring pattern in my development and experimentation was that being extremely explicit and precise in my expectations improved the results drastically.

For each task in the crew, I passed as context all previous tasks. This was important since some information is only stored in one task. For example, the original research question is only stored in the expand task and the source passages are only in the search task.

As I developed the agentic system, I noticed certain tendencies from the agents. For example, the report writer would leave notes to me on the final report, one time it included a “hypothetical link” to nothing just in case I wanted to add the link. In other words, the agent would decide to add components not found in any of the instructions. To address this, I began specifically instructing it not to add notes, hypothetical information, or unnecessary labels.

Throughout this entire development, debugging, and iteration process, I heavily relied on the ReAct style reasoning which I could carefully inspect and review in the terminal. For instance the query expander agent clearly explained its research strategy saying things like “To effectively answer the research question, "What are the key challenges in making AI agents safe and aligned?", we can break it down into the following sub-questions. Each sub-question focuses on specific aspects that will collectively provide a comprehensive understanding of the challenges involved. Relevant keywords and potential search paths are also provided to facilitate in-depth research.” I also found it helpful to review the synthesizer's thoughts as this is what helped me realize that I was not directing it to address the main research question clearly enough. This allowed me to revise the goal for this agent, adding the following line: “Produce meaningful insights based on the reviewed sources and tie your findings to the research question.” It was also helpful to see when the system relied on its memory through the explicit tool calls versus when it decided to search the sources. This led me to experimenting with clearing the memory to see how it affected the results. I found that if the system fell into a pattern of producing an incorrect result (most commonly improper citations), resetting the memory and rerunning the system would often fix this.

I also experimented with increasing the model from gpt-4o-mini to gpt-54-mini and I found that increasing the model tier substantially improved the performance of the system. Gpt-54-mini produced much higher quality synthesis and analysis, fewer formatting issues, consistent citation quality, and better articulated findings. The outputs I included in my submission are from gpt-4o-mini, however if this were to be used in a production system, I think it would be worth spending time investigating which model produces the highest quality results. Additionally, if this were to be used in production, I would like to experiment with incorporating a fifth “reviewer” agent to correct some of the issues produced by the report writer (namely citations), and to ensure the research question is adequately answered. Last, a production system would need either another tool to search for additional documents or an external system that could periodically update the corpus as well as a remote location to store the crew’s memory.